

1 Problem

- LoRA shrinks trainable state, but a 16-bit 65B base plus activations still makes fine-tuning inaccessible.
- Regular 16-bit LLaMA-65B fine-tuning requires over 780 GB of GPU memory.
- Goal: preserve 16-bit quality while storing the frozen model in 4 bits and training only adapters.

2 QLoRA path

- Store pretrained weights in 4-bit NF4; dequantize blocks to bfloat16 only when used in matrix multiplication.
- Gradients pass through the frozen quantized weights into bfloat16 LoRA factors; the base itself is never updated.
- Because the base dominates memory, broader LoRA coverage can be used with little additional footprint.

3 Three innovations

NF4 + DQ + PAGING

- NF4 places equal-probability bins under a zero-centered normal distribution, matching pretrained weight statistics.
- Double Quantization quantizes block scaling constants, saving about 0.373 bits/parameter—roughly 3 GB at 65B.
- Paged optimizers use unified CPU/GPU memory to absorb gradient-checkpointing spikes and avoid out-of-memory failures.

4 Study design

SCALE + DATA + EVALUATION

- Over 1,000 experiments cover 80M-65B models, multiple architectures, instruction datasets and quantization types.
- NF4 with double quantization matches bfloat16 LoRA on MMLU from 7B through 65B and outperforms FP4.
- The study compares human, GPT-4 and benchmark evaluation, showing task scores depend strongly on data-benchmark alignment.

5 Evidence

GUANACO ON ONE GPU

- Fine-tune 65B on a single 48 GB GPU instead of more than 780 GB for regular 16-bit training.
- Guanaco-65B reaches 99.3% of ChatGPT's Vicuna score after 24 hours; Guanaco-33B reaches 97.8% in under 12 hours on 24 GB.
- The 7B model uses about 5 GB at deployment and reaches 87.0% of ChatGPT on the reported Vicuna evaluation.

6 Limits & impact

ACCESSIBILITY WITH CAVEATS

- Full 16-bit fine-tuning equivalence was not tested at 33B/65B because of cost; only LoRA equivalence was established there.
- Chatbot benchmarks are narrow and data-sensitive; responsible-AI analysis and alternative bit widths/adapters remain limited.
- QLoRA democratizes adaptation but is dual-use: cheap fine-tuning enables auditing and privacy as well as harmful customization.

QLoRA

QLoRA

Efficient Finetuning of Quantized LLMs | 2023

Backpropagate through a frozen 4-bit quantized language model into LoRA adapters, making 65B-model fine-tuning possible on one 48 GB GPU while recovering 16-bit adapter performance.

4-bit training

NF4

LoRA